Beyond USB Packet Arrival: Frame-Identified Time Transfer to macOS

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Shared-bus RC/DC comparison; clean-bus validation pending

Abstract

A user-level clock can follow an embedded reference without steering the operating system clock or treating USB packet arrival as event phase. Hardware-captured USB frames, independently identified at the host, connect an immutable source map to Mach time and a private chronyd instance. A fixed concurrent 180 s window compares BLE-derived relay-clock (RC) and local-GNSS Direct Connect (DC) paths after publication and host corrections. Each returned 5,760/5,760 valid USB timing records. Corrected-source causal RX RMS was 190.050 ns (5,054 predictions) for RC and 267.448 ns (5,370) for DC. Delivered chrony-minus-own-fine-source RMS was 325.545 ns (145 comparisons) and 344.546 ns (119), respectively; sampled user-clock availability was 147/177 and 119/177. Chrony's separate estimator medians were 31 and 27 ns. The USB bus was shared, not clean; an isolated-controller repeat remains pending. These are comparable methods and a common window, not a controlled causal ranking or independent UTC metrology. Withheld output, rejected hardware samples, incomplete extended trials and unknown PPS/UTC association and endpoint bias remain explicit.

1 Introduction

The useful timestamp is the time of an event, even when the packet describing it arrives later. This study brings an embedded source to a macOS application over USB. A T096 relay clock (RC) follows an upstream BLE source; a separate Direct Connect (DC) image uses local GNSS/PPS with Bluetooth disabled. Hardware-captured USB start-of-frame (SOF) events link each receiver's counter to the Mac's Mach clock. Control transfers carry captured evidence rather than supplying phase through callback times.

The experimental fine UTC estimate feeds one private user-owned chronyd instance per path, whose virtual clock the application reads back. The operating system clock is unchanged. A selected refclock can produce Stratum 1; that hierarchy indicator is not calibrated accuracy. Compatible USB timing devices can be discovered on different ports. Frame identity on a direct full-speed XHCI connection does not establish that the controller or bus is isolated from other work.

Three measurements answer different questions. Corrected-source RX RMS tests causal source-phase prediction against hardware SOF/Mach boundaries before chrony, excluding the UTC translation. Chrony's tracking RMS describes its internal estimator. A common-instant delivered-clock-minus-fine-estimate comparison measures agreement with the application's own source. We report each population and sampled availability separately. Valid USB replies alone do not establish delivered output.

Earlier revisions corrected repeated coarse-to-fine UTC translation noise, EP0 completion handling and source/map classification. The latest common publisher preserves unchanged-map observations across owner polls, while real authority changes still fence them. This edition replaces the October 15 main result with the October 17 shared-bus window; earlier runs retain their own identities and limitations. No completed clean-bus or two-hour acceptance result is inferred.

2 Related work and comparison criteria

2.1 gPTP, PTP and the Mac clock endpoint

IEEE 802.1AS specifies timing and synchronization for time-sensitive networks, using IEEE 1588 mechanisms where applicable. The IEEE working group's published list inspected on 15 October includes 802.1AS-2025; historical evaluations below use their stated earlier editions [2]. gPTP supplies a network timing architecture with synchronization messages, rate relationships and link-delay handling. The USB prototype in this paper does not implement its message set, peer-delay machinery, election or conformance requirements. The conceptual connection is the transport of event timestamps and the need to cross clock domains without treating software delivery delay as event phase.

A particularly relevant implementation is Clark's *mac-gptp*. It exposes a Mac's AVB gPTP virtual clock through the private TimeSync framework and can feed a chrony reference-clock socket. Its documented setup uses an AVB-capable Ethernet path and a Linux boundary clock. The README reports roughly 250 ns agreement between successive Mach/gPTP mapping estimates over a few seconds and explicitly retains uncertainty in a fixed adapter-to-Mach cross-timestamp offset [3]. This is closely related to our host-endpoint problem, despite the different transport and API.

The project's refclock account reports 0.6 μs RMS over half an hour with tuned chrony settings and approximately 2 μs excursions during its fastest wander. It identifies oscillator wander by comparing the Mac mapping with a separate adapter-clock mapping and explains the importance of short frequency-estimation histories [4]. These are author-reported system-clock-to-gPTP residuals, not our measurements and not a calibrated UTC benchmark. We did not run that project, configure AVB, or steer the system clock. Its distinct adapter clock also provides an attribution experiment that our USB run lacks: our relative rate changes cannot yet be assigned uniquely to the Mac, USB reference, the embedded receiver or its upstream source.

Val et al. evaluate an FPGA-based Ethernet/wireless TSN clock-domain translation with an external Calnex measurement. Their Table III reports -4.3 ns mean and 18.7 ns standard deviation for a three-hop Ethernet setup, and -5.5 ns mean and 16.6 ns standard deviation for one 802.11 plus two Ethernet hops [5]. These are externally observed synchronization distributions in their implementation, not residuals after independently fitting short intervals. The numerical similarity of any short-window figure in our study would not establish an equivalent result.

NIST's calibration study illustrates the missing metrology more directly. Technical Note 2030 reports a 36 ns maximum offset through two transparent clocks with a 6 ns one-sigma uncertainty, and separately discusses source calibration and a power-related anomaly that increased receiver-calibration uncertainty to 100 ns [6]. This is evidence for the importance of reference calibration, not a performance baseline that our experiment can claim to meet.

2.2 USB synchronization and causal delay bounds

Lim, Rosdi and Yap use USB 3.0 Isochronous Timestamp Packets to coordinate device triggering. Their publisher abstract reports approximately 200 ns trigger delay and operation through up to three consecutive hubs [7]. The endpoint is relative triggering among SuperSpeed devices, rather than a full-speed device-to-Mach clock association. It is nevertheless clear prior evidence that USB timing can be much finer than a software transaction interval. Patent literature also describes recovering and synthesizing device clocks from periodic USB structures [8]; we do not present SOF-based synchronization as a new general principle.

WALT's primary clock-synchronization notes develop causal upper and lower clock-offset bounds from messages in both directions. They report interval widths around 100 μs on one Linux USB 2 path and 800 μs on a USB 3 path [9]. Those are bound widths on named configurations, not standard deviations or physical accuracy floors. We use a related causal principle for *integer identity*, while hardware boundary observations supply the

fine phase evidence. Our periodic UTC request intervals deliberately remain wide rather than pretending that a request midpoint is an exact event time.

NTP is a separate network clock-discipline system. RFC 5905 discusses capabilities extending to tens of microseconds on suitable workstations and fast LANs [10]. The historical Internet pool checks do not test that regime. A noisy network observation is not evidence that the underlying NTP-disciplined clock is noisy by the same amount.

3 Timing architecture and coordinate contracts

Let C be the receiver's free-running counter, $M_g(C)$ the source-monotonic projection for immutable map generation$g$, and O its retained integer UTC translation. The source coordinate is

$$U(C) = M_g(C) + O. \tag{1}$$

Let A denote Mach absolute time, K Mach continuous time, and W local macOS wall time. These coordinates have separate meanings and failure boundaries. The USB bridge estimates $A = F(C)$ from hardware SOFs. To evaluate source time at a current host event, the client uses the inverse causal map $C = F^{-1}(A)$ and applies the exact eligible M_g and O. A bracketed association relates A to$K$; local W is used to deliver the SOCK sample and interpret the daemon's clock, not to train the SOF fit.

3.1 Hardware capture and transport

Each T096 uses an nRF52840 TIMER3 at 16 MHz (62.5 ns ticks). A USBD SOF event triggers PPI capture into CC3, so a delayed userspace callback cannot move the original hardware timestamp. Nordic documents a synchronization stage before PPI; timer resolution does not calibrate that stage or the physical USB path[11]. Capture records preserve the timer, frame ordinal, scope and epoch. Acceptance requires coherent frame sampling, strictly advancing sequence/counter, a consistent frame increment and capture age below 16,000 ticks (one full-speed frame).

Vendor EP0 requests retrieve precision SOF/source records at 32 Hz and periodic source information at 1 Hz. The tested interface is FF/54/01, not a standard USB time class. UTS1 kind 1 uses request 0x53; kind 2 uses 0x52. Marker OUT 0x54 and immutable fetch IN 0x55 carry a nonce-bound 64-byte marker tuple. The helper uses the public IOUSBHost APIs, including `referenceMicroframeWithTime:error:`; runtime Mach conversion is 125/3 ns per tick on this setup[12, 13]. The 1 ms frame cadence and 125μs microframe numbering are event spacings, not accuracy floors for timestamped edges.

3.2 Identify the frame before fitting phase

A smooth affine fit cannot distinguish two host/device associations separated by a whole frame. The implementation therefore constrains the integer host-microframe origin using causal zero-data OUT marker transactions and separately recorded host frame-layout checks. Request and completion brackets bound when the marker could have been observed; the marker fetch retrieves the immutable captured tuple later. On this direct full-speed XHCI topology, the intersection of these bounds identifies one integer origin. A 1μs model allowance is part of this identification procedure; it is not an independently calibrated total uncertainty bound.

The bridge requires at least eight valid layout checks and sixteen causal markers, verifies the omission of the three microframe bits in the full-frame API, and retains expiry and consistency fences. A frame-origin change, boot change, USB/capture epoch change or invalid host association invalidates the affected model. Matching this identity is distinct from estimating fine phase. Sleep/resume, hubs and alternate controllers have not been qualified by this experiment.

3.3 Fit a clock using its past

The shipping RX metric retains at most 1,024 accepted source-phase/SOF tuples, approximately 32 s at the requested rate, with at least 64 prior points before forecasting. A frozen previous affine fit predicts the next corrected source-monotonic phase against the hardware SOF/Mach boundary. The current sample enters training only after its prediction and existing hardware and exact immutable-map checks. The UTC translation is excluded. A rejected observation or training gap can reset training, so warm-up can recur inside a settled window.

For signed prediction errors e_i, we report

$$\bar{e} = \frac{1}{N}\sum_i e_i, \qquad \mathrm{RMS} = \sqrt{\frac{1}{N}\sum_i e_i^2}, \qquad \sigma = \sqrt{\frac{1}{N}\sum_i (e_i - \bar{e})^2}. \tag{2}$$

RMS retains bias and transients. A whole-run detrend, rolling fitted residual, base-counter inverse forecast and corrected-source RX prediction are different populations. The October 15 base-USB inverse forecast is therefore not a baseline for claiming a causal improvement from the new RX value. No request midpoint, local-wall residual or future tuple trains the fine bridge. Replay processes all preceding captured observations before scoring the fixed comparison window.

3.4 Source permission remains original

Each usable source record carries the exact map, full root/boot/source/path scope and original permission deadline. The client evaluates whether the captured or projected counter lies in that map's interval and clips host use to the original deadline. A later map cannot retroactively authorize an unavailable event. Source restart, interpretation change, invalid UTC, malformed input, unknown classifications and original expiry retain their withdrawal fences.

A missing historical-SOF geometry classification remains an invalid time sample: flags and captured UTC are zero. The host withholds output and cancels pending sends, retaining private training only when the current counter remains eligible under the exact old map and unchanged original deadline. True withdrawal and expiry discard relevant state. The common USB publisher now treats unchanged owner polls idempotently, without replacing an in-flight observation. Permission, scope or session-shortening changes still fence it. RC USB export no longer requires BLE relay transmit ownership; this changes neither BLE discipline nor the GM/RC dispatcher. DC remains a separate flash-only, Bluetooth-disabled image, outside the runtime GM/RC switch.

4 User-level chrony integration

The normal CLI starts one owned private chronyd 4.9 instance with `-x -U -u CURRENT_USER`, a private command socket and a SOCK refclock. The documented `-x` mode disables system-clock adjustment[15]. The local manager uses a contained known configuration, waits for both sockets, and reaps only its own child. No administrator service, system timing configuration or direct Mac Bluetooth receiver is involved. Compatible relay discovery spans USB ports; identity checks and exclusive ownership prevent accidental mixed-device records.

The default fine feed composes the causal SOF/Mach bridge, the immutable source projection and the retained UTC translation at a common host instant. Integer UTC differences avoid epoch-sized floating-point subtraction. Each admitted fine observation retains a wall/Mach bracket no wider than 2μs, its original grant deadline, source generation and packet provenance. This admission limit does not become a 2μs absolute accuracy claim. A generation fence cancels pending feeds on authority transitions.

The application monitors chrony's tracking correction and rate to project its user-level clock in local wall coordinates, subject to Mach freshness and wall-step guards. Both current paths use the quarter-second SOCK configuration `poll -2 filter 4 precision 0.05 delay 0.2`, with `minsamples 4` and `maxsamples 64`.

The 50 ms precision and 200 ms delay are provisional configured allowances, not measured uncertainty bounds or tracking RMS. The SOCK refclock has stratum 0; the observed selected virtual clock reports stratum 1. No forced local stratum, fabricated PPS, trust override or network fallback creates a successful selection. Tracking RMS is the daemon's historical offset statistic[15]; the UI labels it CHRONY EST RMS and uses whole nanoseconds below 1μs.

Presentation and authority remain separate. For a display glitch strictly shorter than 500 ms, the large clock/RMS hold the last visible values with an immediate loss indicator; at 500 ms the outage presentation applies. Held pixels never feed chrony or serve clock APIs. The chrony page exposes source, frequency, freshness, filter/feed and lifecycle statistics. A smooth screen is not evidence of continuous time availability.

5 Experiment and fixed-window analysis

The selected window spans approximately 12:42:10–12:45:10 CDT on 17 October 2026 (17:42:10–17:45:10 UTC), exactly 180 s in common host-monotonic coordinates. It covers RC trial +60 to +240 s, selected by duration and concurrency rather than error magnitude. Both paths used the same Apple silicon Mac, macOS 26.7 build 25G229, chrony 4.9, 32 Hz USB requests, `poll -2`, `filter 4`, `maxsamples 64` and shipping RX arithmetic. RC firmware was 0.15.81 at `6c6f1b27752e5f31906e8de08bc148205699c045`; DC was 0.15.80 at `908e11a7da2d6d36053be31fc37e6c63224f6cf2` [14].

The paths were GNSS–BLE–RC–USB–macOS and separate local GNSS–DC–USB–macOS. They did not share a physical GNSS source chain. RF/GNSS placement, temperature and host load were not controlled treatments. **The USB bus was shared, not clean**: other USB devices and concurrent timing work were present. A matched physical repeat on an isolated controller remains pending. Direct-port frame association is not a clean-bus qualification.

Every row in the fixed window remains in its population. Hardware validation and RX warm-up are separate from valid replies. Native approximately 1 Hz statuses supply delivery denominators and sample the latest synchronized tracking readout; no missing output is filled with zero error. Exact comparisons use separate admission brackets, so delivery and comparison counts can differ. Point samples do not measure continuous outage duration.

The direct comparison computes delivered chrony UTC minus its fine estimate at the same admitted host wall/Mach instant, using integer differences. Source or endpoint error common to both may cancel. It is distinct from chrony's reported RMS and is not independent UTC truth. No operating-system clock adjustment or network serving was configured. The requested two-hour trials were still running when this window was selected; neither their completion nor long-run durability is claimed.

6 Results

6.1 USB records and corrected-source RX prediction

Table 1 retains the complete record ledger. Both paths returned all 5,760 expected timing replies as valid precision records, with no unavailable or missing reply. Hardware acceptance is a further condition: 12 RC and six DC observations were rejected. RX forecasts scored 5,054 and 5,370 predictions; the remaining accepted observations were warm-up. No unqualified RX observation or source-map mismatch was recorded, and each path had one source scope.

Each path's predictions, warm-up observations and hardware rejections sum to all 5,760 replies. Separate source chains, finite populations and uncontrolled bus/load conditions prevent a causal RC/DC ranking.

Table 1: Fixed 180 s window on a shared, not clean USB bus. RX excludes UTC translation; predictions precede training.

Measurement	RC	DC
Valid / expected USB timing replies	5,760 /5,760	5,760 /5,760
Unavailable / missing precision replies	0 /0	0 /0
Hardware accepted / rejected	5,748 /12	5,754 /6
RX predictions / warm-up	5,054 /694	5,370 /384
RX unqualified observations	0	0
RX RMS (ns)	190.050	267.448
RX mean / population σ (ns)	6.455 /189.940	18.896 /266.780
Map mismatches / source scopes	0 /1	0 /1

6.2 Chrony estimator, delivered agreement and availability

Table 2 separates status-sampled tracking readouts, direct comparisons and delivery. Each path had 177 synchronized readouts, with median RMS of 31 ns for RC and 27 ns for DC. These cached internal statistics are not independent accuracy trials. Delivered chrony-minus-own-fine-source RMS was 325.545 ns across 145 RC comparisons and 344.546 ns across 119 DC comparisons.

Table 2: Separate chrony populations in the same shared-bus window. All differences are relative to each path's own fine source.

Statistic / population	RC	DC
Status-sampled tracking readouts	177	177
Estimator RMS: minimum / median / maximum (ns)	14 /31 /109	14 /27 /116
Exact common-instant comparisons	145	119
Chrony-minus-fine RMS / mean (ns)	325.545 /−69.379	344.546 /−85.437
Absolute P99 / maximum (ns)	837 /909	689 /831
User-clock status: delivered / withheld	147 /30	119 /58
All native status observations	177	177
Transport errors / source-fence events	0 /0	0 /0
Chrony daemon epochs in window	1	1

Absolute P99 uses nearest rank $\lceil 0.99N \rceil$. All admitted direct differences in this window were below 1μs, but withheld observations have no measured error and cannot inherit that bound. User-clock availability was 147/177 for RC and 119/177 for DC, even though every USB timing reply was valid. RC had two more delivered statuses than exact comparisons because those observations use separate sampled brackets. These denominators describe sampled delivery, not continuous service.

There were zero recorded USB transport errors or source-fence events, one daemon epoch and one source scope per path. This supports stable operation during the selected window; it does not prove the absence of failures before or after it. No claim that the smaller estimator median predicts smaller delivered-clock error is supported: the estimator and direct comparison measure different things.

7 Corrections and separate historical evidence

7.1 What was corrected

Earlier publication recomputed an integer UTC translation from coarse/raw UTC while the precise monotonic map evolved independently. Binding one coherent translation removed repeated millisecond variation in the

October 15 corrected run. Contradictory associations and restart still fence replacement. A constant translation can retain unknown initial PPS/RMC second association and fixed bias; the correction does not commission a UTC anchor.

Subsequent EP0, hardware-SOF coherence and publication fixes distinguish transport, accepted capture and source permission. Firmware 0.15.81 consolidates DC's unchanged-map rule into the common publisher: an unchanged RC poll no longer races an in-flight serialized record. Real permission/scope changes and shortened session deadlines still revoke it. Receiver-only RC USB no longer depends on owning BLE relay transmission. Source discipline, RF behavior and GM/RC dispatch remain unchanged; DC is still flash-only with Bluetooth disabled.

Clean GM/RC/DC builds and image inspections, identical GM/RC executable application sections, role-manager and watchdog/USB sanitizer checks, and 8,808 macOS native checks support software behavior. They are software and short functional evidence, not UTC metrology or a controlled attribution of the new hardware statistics to one fix. The concurrent window was not a before/after treatment experiment.

7.2 Failed acquisition and earlier populations remain separate

The first RC capture after readback/reboot had no usable UTC for 60 s, with delayed BLE reacquisition at approximately 93 s. It remains a separate failed acquisition population, not pooled into this settled window or excused as successful delivery. Startup/reacquisition behavior remains under investigation. The ongoing extended trials are incomplete evidence.

The October 15 final 300.065 s experiment, using firmware 0.15.53 and CLI 0.21, had 9,437 qualified base-counter inverse forecasts at 328.413 ns RMS. Its predeclared 60–300 s chrony estimator median was 129 ns across 265 queries; direct agreement was 414.351 ns RMS with 182 valid and 54 withheld of 236 statuses. The 32.072201 ms startup estimator transient remains in that collection. Zero USB errors and zero actual daemon restarts described that finite run, not all later operation. Base-counter forecasts and current corrected-source RX forecasts are different methods and are not pooled.

Its preceding .52/CLI.20 run recorded 46 actual chrony restarts and 103 valid/133 withheld settled comparisons, despite zero USB errors and 414.137 ns direct RMS. Earlier 600 s exploratory block predictions had 1.053μs RMS, while retrospective short-window phase was quieter; this did not establish a causal full-run output bound. Historical millisecond UTC noise, source gaps, rejections and failed attempts remain preserved. Their correction does not erase those populations or qualify later builds.

8 Limits and next validation

The current shared-bus window establishes sub-microsecond relative RX and delivered agreement RMS on two concurrent paths, with explicit rejection and delivery denominators. It demonstrates user-level chrony without system-clock control. It does not measure independent UTC error. Shared-source error can cancel; unique frame identity, 62.5 ns timer ticks and a narrow admission bracket are not a complete uncertainty budget. PPS-to-UTC association and Apple/controller/PHY/cable-to-Mach bias remain unknown.

The next physical validation is a matched isolated-controller repeat on a **clean USB bus**. Record controller topology, competing devices and load, retain the same windows, causal replay, admission rules and denominators, and report bias, RMS, tails and withholding together. Shared and clean-bus cohorts must remain separately identified. Until that repeat is complete, this paper makes no clean-bus acceptance or causal bus-load claim.

Complete the extended trials and account for acquisition and every output refusal, including an explicit gate reason and admission generation. Valid transport with withheld output remains an availability problem even when admitted differences are small. Preserve reset, wall-step, source-transition and expiry fences while investigating

it. Adaptation or tuning requires fresh frozen causal validation; retrospective fitted residuals cannot substitute for delivered output.

Independently commission PPS/UTC labeling and measure a second timed host event or external reference to expose common-mode endpoint error. Matched USB, NTP and gPTP comparisons require a common independently timed endpoint, horizons and availability. There is no qualified alternate-Mac, hub, suspend/reconnect, thermal or CPU/I/O stress cohort, leap-second result or adversarial-source evaluation. Exclusive ownership prevents accidental mixing, not malicious USB impersonation. Continuous precision, absolute traceability, long-run durability and transport superiority remain unqualified.

9 Conclusion

After publication and client corrections, the fixed 180 s RC/DC window returned all USB timing replies and showed corrected-source RX RMS of 190.050/267.448 ns. Delivered chrony agreement with each path's own fine source was 325.545/344.546 ns RMS, while sampled user-clock availability was 147/177 and 119/177. Separate chrony estimator medians were 31/27 ns. These shared-bus observations retain hardware rejections and withheld output; they neither rank the source paths causally nor establish UTC accuracy. Clean-bus repetition, completed durability trials and independent endpoint calibration remain necessary.

Data, code and reproducibility

This research draft is not submitted or accepted. Jonathan Yantis's selected summary, `clients/macos/results/2026-10-17-ble-vs-direct.json`, binds the window, software, raw-prefix identities, metrics and exclusions. Sanitized observations at `evidence/usb/2026-10-17/` permit summary recomputation; private raw captures, enrollment and device bindings remain excluded. Physical reconstruction still requires those captures. A separate first-300 s shipping-replay cross-check reproduced 8,364 predictions and 188.215650 ns RX RMS; it validates arithmetic, not the comparison window.

The standalone source and matching PDF are bound by `papers/editions/usb-macos-2026-10-17.json`. Git preserves the prior edition, whose immutable import receipt and historical CSVs remain under `papers/imports/research-2026-10-16.json` and `evidence/research/2026-10-16/`. The public offline command `python3 tools/verify_research.py` checks the retained historical subset and current edition identities/accounting without operating hardware. It does not recreate excluded raw captures or independently qualify UTC. Original paper and figures: copyright 2026 Jonathan Yantis, CC BY 4.0.

Author contributions and assistance disclosure

Jonathan Yantis supplied hardware, protocol/firmware, experimental direction and research decisions. OpenAI Codex assisted implementation, tooling, literature research, evidence review, analysis, writing and LaTeX preparation, including this revision's checks against the selected results and shipping arithmetic. The human author retains responsibility for evidence, claims, references, privacy and venue requirements. No AI system is a human coauthor; no affiliation, acceptance, DOI or arXiv identifier is claimed.

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